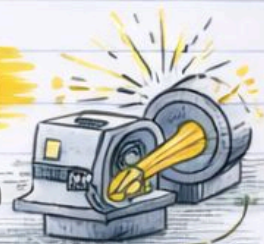


Manufacturing Machining

Module-4 Notes

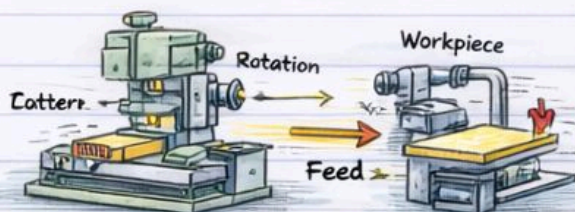
by pyqfort.com

Grinding Wheel

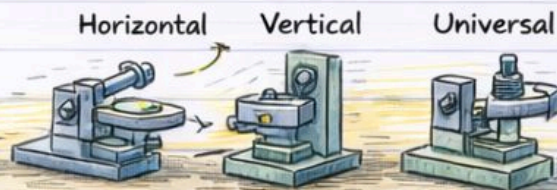


Contents Covered:

• Milling and it's Principle



• Milling Machines

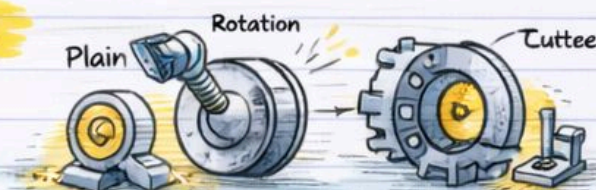


• Milling Operations

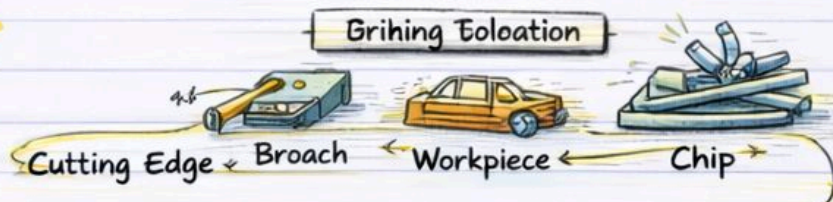


• Types of Milling Cutters

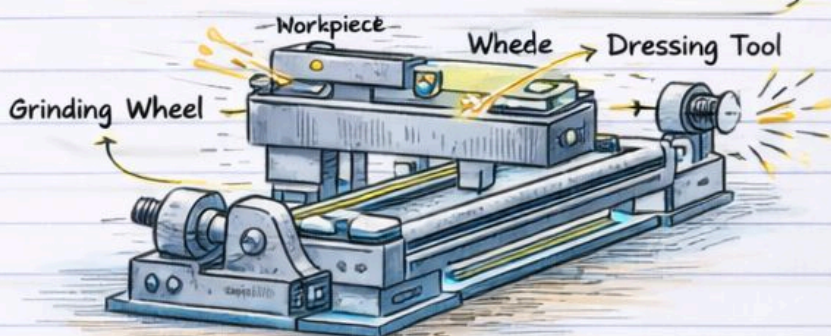
• Grinding and it's Principle



• Chip formation



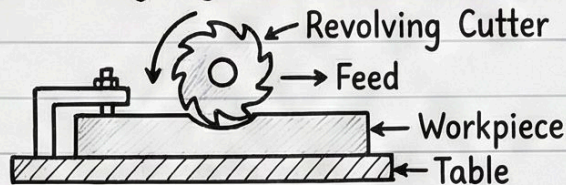
• Redressing



Milling and its Principle

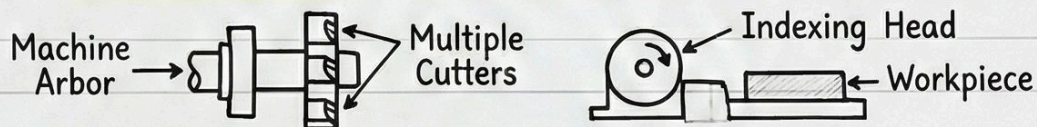
1. What is Milling?

- Milling is a machining process in which metal is removed by means of a revolving cutter with many teeth.
- The workpiece is supported on a table and may be fed to the cutter, longitudinally, transversally or vertically.
- It is very useful for producing cylindrical surfaces and especially plane surfaces.



2. Importance & Versatility:

- The milling machine is the most versatile machine tool, used for nearly all operations with high accuracy and surface finish.
- Can produce a wide variety of flat and formed surface.
- Use of indexing head and possibility of mounting several cutters on the machine arbor can increase metal removal rate.



3. Principle of Milling Machine:

- Flat as well as curved surfaces are generated by rotating multi-edged cutting tool called milling cutters.
- The milling cutter is rotated and the feed is given to the workpiece.
- Main milling processes: slab milling, face milling, slot milling, form milling, angular milling.

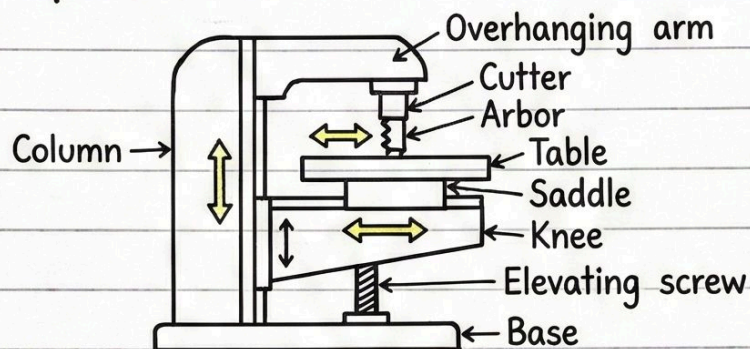


Fig. 4.36

Column and Knee Type Milling Machines

1. General Description:

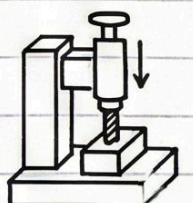
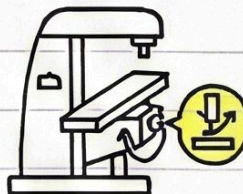
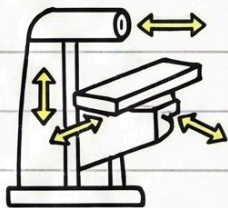
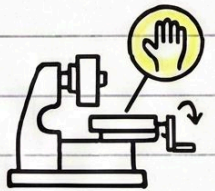
- This type is the most commonly used in production shop.
- The work table is mounted on a knee casting which can slide up and down on a hollow column.
- These machines are further classified as per method of power supply to the table, different axis of rotation, and different movements of the table.



(Based on Fig. 4.37)

2. Types:

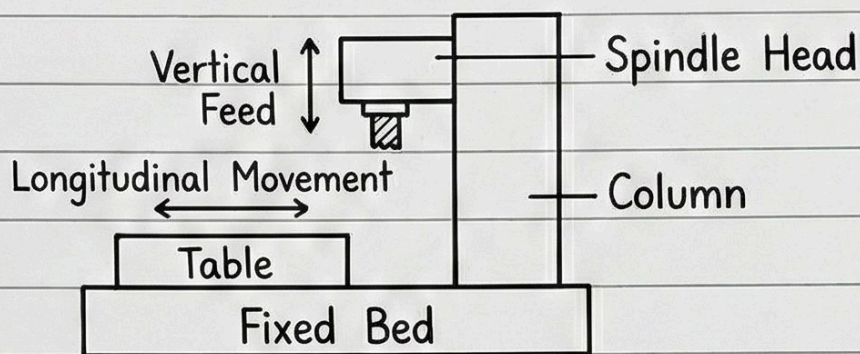
- Hand Milling Machine**: Simple and small for light milling operation (keyways, grooves, slots). Manual feeding of workpiece.
- Plain Milling Machine**: Rigid and sturdy for heavy jobwork. Horizontal spindle, table fed manually or by power in vertical, cross or longitudinal direction.
- Universal Milling Machine**: Tool room machine for very accurate jobs. Table mounted on a circular swivelling base. Can produce gears, drills, reamers, etc.
- Omniversal Milling Machine**: Tool room machine with all universal features plus a special knee assembly for longitudinal direction horizontally. Can machine taper spiral grooves.
- Vertical Milling Machine**: Cutter is mounted on a vertical spindle. Can be plain or universal type. End mills and face milling cutters used for grooves, slots, flat surfaces.



Fixed Bed Type Milling Machine

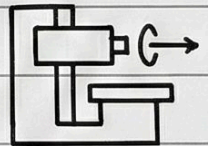
1. General Description:

- These are high production milling machines.
- The work table is mounted on a fixed bed and has only longitudinal movement.
- The adjustable spindle head is attached to a vertical column and can be fed vertically.

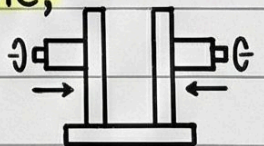


2. Special Features/Types:

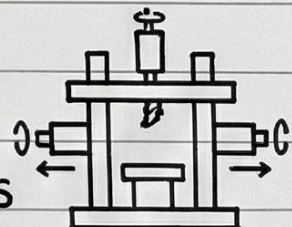
- (i) Simplex Milling Machine: Has single adjustable, horizontal spindle head.



- (ii) Duplex Milling Machine: Has two adjustable, horizontal spindle heads attached to two separate vertical columns, one on each side of fixed bed.



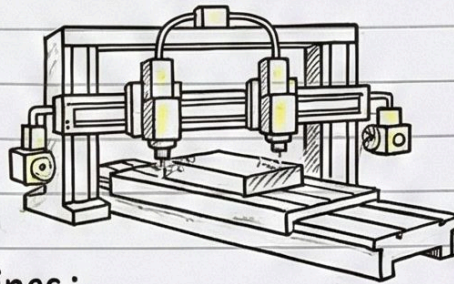
- (iii) Triplex Milling Machine: Has two adjustable, horizontal spindle heads attached to two separate vertical columns and additional vertical spindle head mounted on a cross rail.



Planer and Special Milling Machines

1. Planer Milling Machine:

- It is a heavy duty production machine which is similar to double housing planer.
- Spindle carriers are mounted in place of planer tool posts and are operated by individual motors.
- Two additional vertical milling heads are fitted on a cross rail and can travel along it.

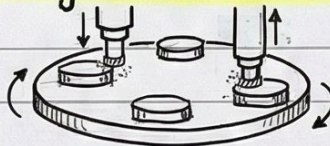


2. Special Milling Machines:

- Designed for specific jobs, having unique spindle and workpiece/cutter movement arrangements.

(i) Rotary table machine:

- A mass production machine with a heavy, rotating, circular work table.
- Has two vertical spindles with facing mill cutters set at different heights for roughing and finishing.
- Allows continuous loading and unloading of workpieces.



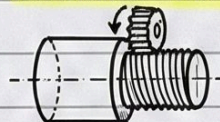
(ii) Drum milling machine:

- Workpieces are mounted on the periphery of a rotating drum fixture against rotating cutters (up to four).
- Operation is continuous and suitable for mass products.



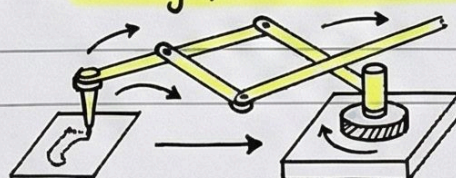
(iii) Planetary milling machine:

- Revolving cutters move in a planetary path to finish a stationary cylindrical surface (internal/external).
- Specially used for cutting internal or external threads of different pitches.

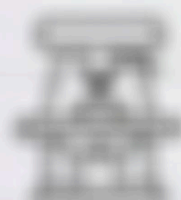


(iv) Pantograph milling machine:

- Used for duplicating and profiling work like die and mould cutting, grooving.
- Tool position is guided by a templet of the model.
- Pantograph helps to enlarge, reduce or copy the same size.



Milling Operations



1. Introduction:

- The **main operations** which are carried out on a **milling machine** are listed below

2. Common Milling Operations:

(i) **Plan or Slab Milling:** Produces flat surface parallel to the axis of the cutter.



(ii) **Face Milling:** Produces flat surface perpendicular to the axis of the cutter.



(iii) **End Milling:** Used for producing flat surfaces, slots, grooves, and keyways.



(iv) **Side Milling:** Produces flat surfaces on the sides of the workpiece.



(v) **Slot Milling:** Used for cutting slots, grooves, and keyways e.g., T-slot, Woodruff key-seat.



(vi) **Angular Milling:** Produces flat surfaces at an angle to the axis of the cutter.



(vii) **Form Milling:** Used for producing irregular or curved surfaces using a formed cutter.



(viii) **Straddle Milling:** Two or more side milling cutters are used to machine two parallel vertical surfaces simultaneously.



(ix) **Gang Milling:** Several milling cutters are mounted on the same arbor to machine several surfaces simultaneously.



(x) **Slitting or Saw Milling:** Used for cutting off workpieces or producing narrow slots.



3. Other Operations:

- Other Important operations include: Keyways, grooves and slots, Profile milling, Cam milling, Thread milling, Gear cutting, Helical milling.

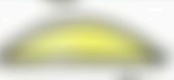
Types of Milling Cutters

Introduction:

- There are many different types of milling cutters. Some are



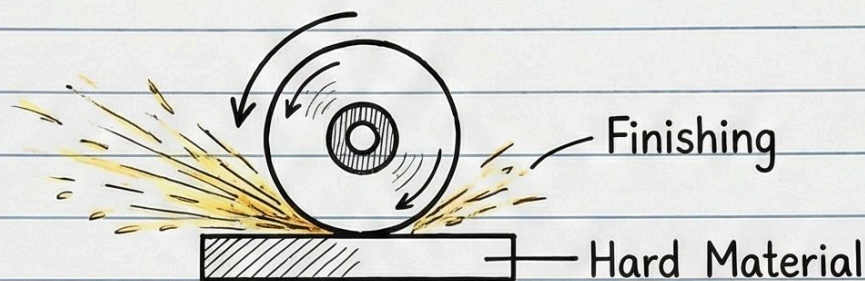
Classification of Cutters:

- Plain milling cutter:** Used for flat surfaces.
Types: (i) Light duty, (ii) Heavy duty, (iii) **Helical** plain milling cutter. 
- Side milling cutter:** Cuts on periphery and sides.
Types: (i) Plain, (ii) Staggered teeth, (iii) Half side, (iv) Interlocking. 
- Metal cutting saw:** For cutting off or narrow slots.
Types: (i) Plain, (ii) Staggered teeth. 
- Angle milling cutter:** For angular surfaces.
Types: (i) Single angle, (ii) Double angle. 
- End mill:** Cuts on end and periphery.
Types: (i) Taper shank, (ii) Straight shank, (iii) Shell end mill. 
- T-slot milling cutter:** For cutting T-slots. 
- Wendroff key slot milling cutter:** For wendroff keyways. 
- Fly cutter:** Single-point tool on a bar. 
- Formed cutter:** For irregular shapes.
Types: (i) Convex, (ii) Concave, (iii) Corner rounding, (iv) Gear, (v) Thread.  
- Tap and reamer cutter:** For making threads and finishing holes. 
- Inserted tooth cutter:** Teeth are separate inserts. 

Grinding and its Principle

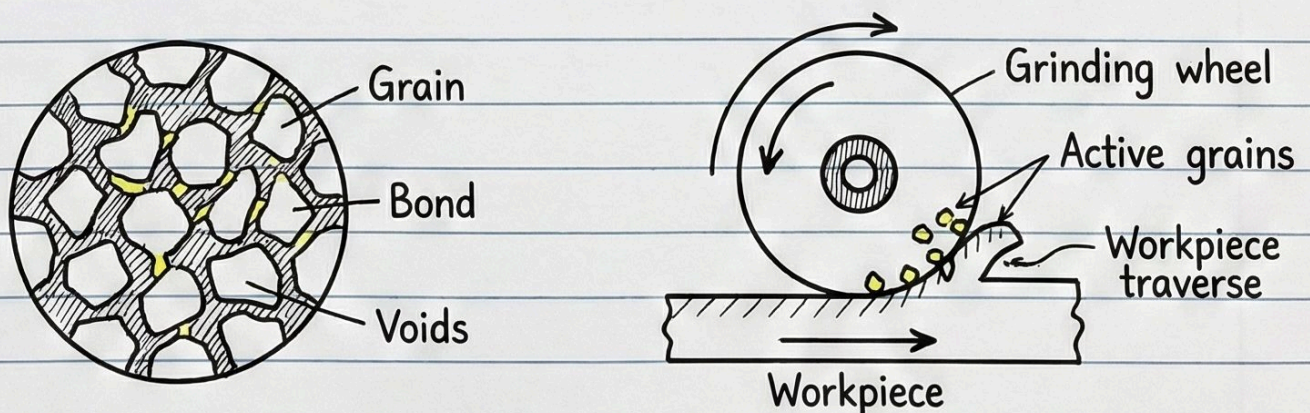
1. What is Grinding?

- Grinding is a metal cutting operation in which a rotating abrasive wheel is used as a tool.
- It is generally used as a finishing process to get high surface quality, correct size and accurate shape.
- Also used for machining hard materials which are difficult to cut by other tools.
- New processes like high speed grinding and creep feed grinding allow for bulk material removal.



2. Principle of Grinding:

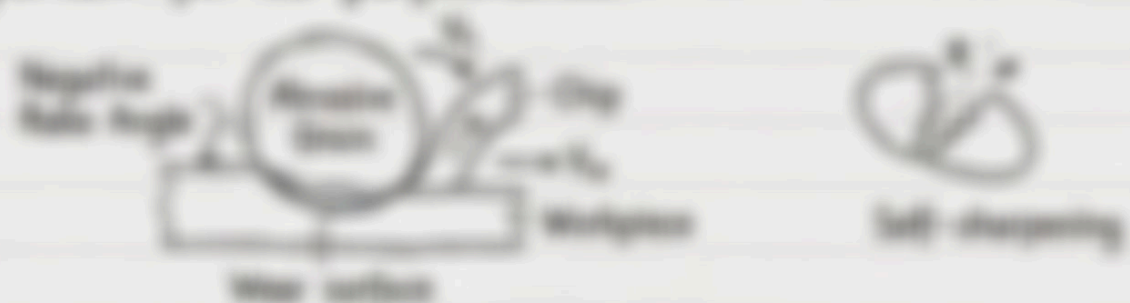
- A grinding wheel consists of abrasive particles called grains held together by a bond.
- The active grains on the wheel surface perform the cutting action like a multi-point tool (e.g., a milling cutter).
- Each irregularly shaped and randomly distributed active grain removes a short chip of gradually increasing thickness.



Chip formation and Redressing

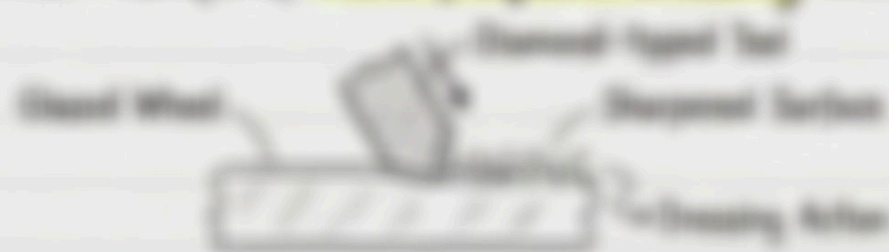
1. Chip formation:

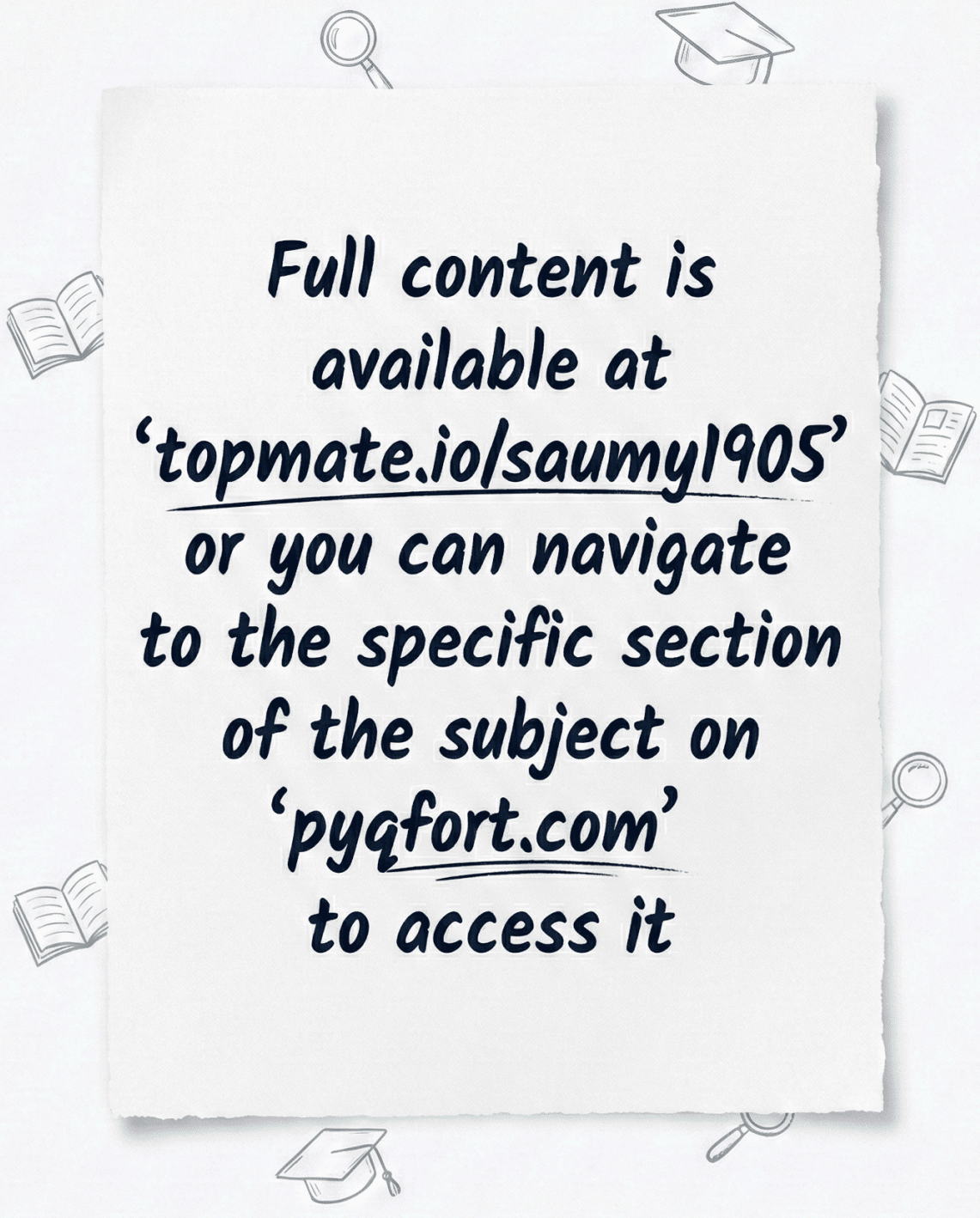
- The active grain is a **small cutting tool** with a **large negative rake angle** (30° or more).
- The **ploughing** action results in **progressive wear** and the formation of **worn areas** on the active grains.
- With the progress of grinding, the size and number of worn areas increase. This results in **increase of friction and force** acting on the grain.
- When the **force becomes very large**, the worn grain is **fractured** and a **new cutting edge** is exposed.
- This **self-sharpening property** of the grinding wheel is very important for its performance.



2. Redressing:

- If the wheel is made of **tough grains bonded together strongly**, the self-sharpening characteristic is **poor** and a **glazed appearance** is seen.
- The grinding wheel of this stage requires **dressing**.
- **Dressing with a diamond-tipped tool** removes the **worn grains** from the wheel periphery and a **new sharpened cutting surface** is generated.
- In case of **soft wheel**, the volume of wheel removed will be significant... Therefore, **more frequent dressing** will be required.





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